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An Unexpected Effect of Ketone Body and GH Co-Administration

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Abbreviations: β -HB, β -hydroxybutyrate; β OHB, sodium-3-hydroxybutyrate; FFA, free fatty acid; GH, growth hormone; HCA2, hydroxy-carboxylic acid; HEC, hyperglycemic-euglycemic clamp; IGF-I, insulin-like growth factor I; SRIF, somatostatin analogue.

The main sources of energy for our cells are carbohydrates and fat. After a regular meal, the main energy source will be glucose. The glycogen stores of the liver and to a lesser extent skeletal muscle can provide glucose as the main energy source for 10 to 14 hours of fasting (1, 2). Thereafter, energy restriction will lead to hydrolysis of triglycerides to free fatty acids (FFAs) by adipocytes and gluconeogenesis by the liver. The hepatocytes then convert the FFA into the ketone bodies, β -hydroxybutyrate (β -HB) and acetoacetate, respectively (1, 2). These ketone bodies can be actively transported into cells and used as an energy source.

During exercise, ketogenic diets, and intermittent fasting, ketone body levels increase (1, 2). Many beneficial effects such as metabolic improvement, preserved muscle mass, anti-inflammation and antiaging have been attributed to the increased levels of ketone bodies (1, 2). These observations indicate that ketone bodies are not only an alternative fuel supply, but can act as signaling molecules as well (1, 2). Although many beneficial effects have not been completely elucidated, some of the more common actions of β -HB as regular “fuel” supplier are not completely clear either. The general concept is that, in rodents at least, either β -HB or glucose is taken up and used in muscle cells (3). However, a recent study by Høgild and colleagues (4), challenges this classic concept.

In this randomized, single-blinded crossover study, 8 healthy lean males were included after an overnight fast. They were infused for 6 hours with (30 ng/kg/min) human recombinant growth hormone (GH), combined with either sodium-3-hydroxybutyrate (β OHB) (0.89 g/kg/h) or saline. After the 3 hours of infusion, a hyperglycemic-euglycemic clamp (HEC) was performed between 180 and 360 minutes. The primary outcome was insulin sensitivity during the HEC (4). β OHB infusion induced about a 100% increase in insulin sensitivity, as defined by the M value and glucose infusion rate (4). This was accompanied by a 20% increase in energy expenditure assessed by indirect calorimetry (4). The GH-induced increase in FFA and

glycerol levels was blocked by β OHB infusion, probably via the inhibitory G protein-coupled hydroxy-carboxylic acid (HCA2) receptor, as suggested by the authors. High levels of FFAs during GH infusion led to insulin resistance, which was blocked by the co-infusion of β OHB. The muscle biopsy, during β OHB infusion, contained high amounts of β OHB and acetoacetate, suggestive of active metabolism of β OHB in the muscle, while endogenous ketone bodies were low both in saline and the β OHB infusion. Glycogen synthesis and storage were unaffected by β OHB, as assessed by polymerase chain reaction and glycogen synthase activity in the muscle biopsies. The authors suggest that the infused glucose during the HEC is metabolized in the muscle at the same time as the infused β OHB. This is in contrast to Randle’s hypothesis that claims that at any given moment either glucose or ketone bodies can be used as fuel source (3). The conclusions by Randle come from observations in rodent tissues from muscle and heart (3), so maybe species-specific differences do apply. In the Danish study the assessment of the metabolism of glucose and β OHB metabolism was mainly indirect, and although it seems plausible a direct assessment would be preferable to prove the concomitant utilization of glucose and β OHB. The individuals in this study were fasting, lean young males and were exposed to a GH excess. Therefore, it is not certain if these results can be extrapolated to people without an overload of GH, females, overweight, or obese individuals. There are indications that without GH infusion during an HEC and with concomitant infusion of lipids, a lower utilization of glucose as energy source can be observed (5).

In the study by Høgild et al (4), GH dependent or not, β OHB administration suppressed FFAs and increased energy expenditure plus improved insulin sensitivity, which are intriguing data that may have clinical implications for specific diseases or conditions. Patients with, for example, acromegaly and a normal insulin-like growth factor I (IGF-I) during long-acting somatostatin analogues (SRIFs) treatment can still have elevated GH (6). Together with

acromegaly patients that have mildly elevated levels both of GH and IGF-I, after surgery or during SRIF treatment, these could be attractive groups to be treated with a ketogenic diet. The increased ketone bodies and elevated GH levels may be an interesting combination. In a pilot study by Coopmans et al (7), acromegaly patients with moderate disease activity had a decrease in IGF-I levels in response to a eucaloric ketogenic diet. This pilot study by Coopmans and colleagues did not assess metabolic parameters that could be of potential interest when thinking about the observations by Høgild et al. Nevertheless, beneficial effects of ketone bodies during acromegaly treatment might be limited, since there are no data on direct or indirect effects of SRIFs, pegvisomant, or dopamine agonists on the HCA2 receptor. Especially since Høgild and colleagues suggest that this receptor is responsible for the decrease in FFAs during β OHB infusion, more studies into this are mandatory.

Other metabolic diseases like obesity and type 2 diabetes mellitus might benefit as well. There are indications that low-dose GH treatment can have some beneficial effects in type 2 diabetes mellitus (8). So maybe low-dose GH administration combined with a ketogenic diet or ketone body administration, thereby blocking an increase in FFAs, could turn out to be beneficial. If the combination of high GH and ketone body levels indeed results in increased energy expenditure, improved insulin sensitivity, and better utilization of glucose, that would be very beneficial for these groups of patients. Additional benefits can emerge from the concept that ketone bodies might act as signaling molecules (1, 2), which hopefully will invite more researchers to dig further into this topic.

Disclosures

The authors have nothing to disclose.

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